Assignment 2: Simple Policy Q&A Bot

1. Introduction

This project implements a retrieval-based Question–Answering (Q&A) system that extracts relevant information from three policy documents: Leave Policy, IT Policy, and Travel Policy. The system uses classical Natural Language Processing (NLP) techniques—specifically TF-IDF vectorization and cosine similarity to match user queries with the most relevant policy statement.

The goal is to build a transparent, efficient, and cost-free solution without relying on paid APIs, large language models, or GPU-based systems.

2. Approach

2.1 Document Parsing and Statement-Level Chunking

Each policy document is parsed and divided into individual numbered statements using regular expressions. Instead of retrieving entire documents, the system operates at the statement level.

This improves precision by returning the exact policy rule relevant to the user's question. For each statement, the system stores:

- A cleaned version (for vectorization),
- The original version (for display),
- The source document name (for transparency).

2.2 TF-IDF Vectorization

All policy statements are converted into numerical vectors using Term Frequency–Inverse Document Frequency (TF-IDF).

- **Term Frequency (TF)** measures how frequently a word appears in a statement.
- **Inverse Document Frequency (IDF)** reduces the weight of words that appear across many statements and increases the importance of rare, distinguishing terms.

This ensures that domain-specific words such as "VPN", "maternity", or "reimbursement" carry higher importance than common words like "is" or "employees".

The output is a TF-IDF matrix where each row represents a policy statement and each column represents a unique term in the vocabulary.

2.3 Query Processing and Similarity Matching

When a user submits a query:

1. The query is pre-processed (converted to lowercase and cleaned).
2. It is transformed into a TF-IDF vector using the previously fitted vectorizer.

3. Cosine similarity is computed between the query vector and all policy statement vectors.
4. The statement with the highest similarity score is selected as the best match.

Cosine similarity measures the angular similarity between vectors and produces a score between 0 and 1. Higher values indicate stronger relevance.

2.4 Threshold-Based Fallback Mechanism

To prevent incorrect answers, a similarity threshold is introduced.
If the highest similarity score is below the threshold, the system returns:

"Information not available in policy documents."

This ensures reliability and avoids misleading responses when no relevant policy statement exists.

3. Key Insights

- **Statement-level retrieval improves precision** by returning exact rules instead of full documents.
- **TF-IDF effectively captures term importance**, highlighting distinguishing policy terms.
- **Cosine similarity enables ranked retrieval** with confidence scoring.
- **The system is transparent**, displaying the retrieved statement, source document, and similarity score.
- The approach is lightweight, interpretable, and suitable for small-to-medium structured document collections.

4. Limitations and Future Scope

Since TF-IDF is a bag-of-words model, it does not understand semantic meaning or synonyms (e.g., "vacation" may not match "leave"). For more advanced applications, semantic embeddings such as Sentence-BERT or LLM-based retrieval methods could improve contextual understanding.

Conclusion

This project demonstrates how classical NLP techniques can be used to build an efficient, interpretable, and cost-effective document retrieval system. Despite its simplicity, the system captures the fundamental principles of modern retrieval-based question-answering pipelines and provides a strong foundation for further enhancement.